ALY 6040 Data Mining Applications

Module 2 – Final Project - Milestone 1 EDA

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Introduction:

1. **Dataset Overview:**

The dataset contains **100,000 entries** and 20 columns. Each entry corresponds to an employee, with columns detailing attributes like department, gender, age, job title, salary, performance, resignation status, and more.

2. **Missing Data and Duplications:**

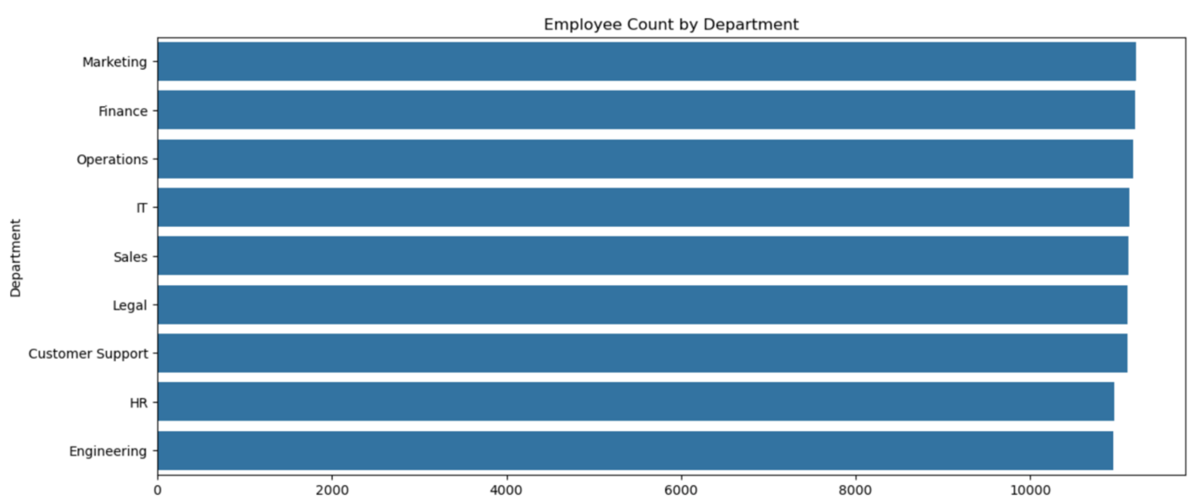
- Upon calling df.info(), no missing data is present in any of the columns. Every column has 100,000 non-null entries.
- Duplicate entries were not explicitly checked in the provided code. However, based on the structure, it would be prudent to run df.duplicated().sum() to identify if there are any duplicated records.

3. **Outliers and Suspicious Data:**

- **Outliers**: The boxplot of monthly salary revealed some potential outliers. While most salaries fall within the range of $4,000–$9,000, there may be some extreme values or inconsistencies in certain job roles or genders that require further investigation.
- Some employees have 0 sick days or 0 overtime hours, even in high workload roles. This might be due to specific job structures (e.g., contractors) or underreporting, and needs validation.
- In the Resigned column, employees with very high overtime or low satisfaction scores should ideally correlate with resignation, but if such patterns don't emerge, that may signal data errors or more complex employee retention dynamics.

4. **Data Visualizations**

i. **Countplot for categorical variables**

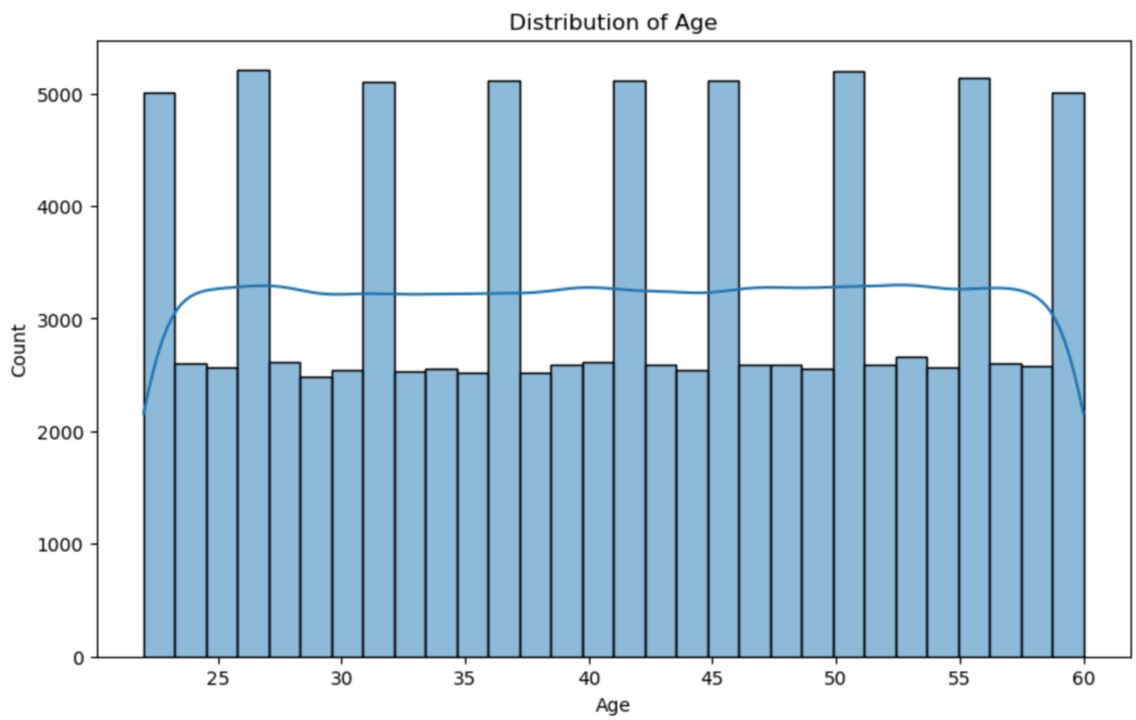


Description:

- Marketing: Has the highest number of employees, with a bar extending close to 10,000.
- Finance: Follows with a significant count, but less than Marketing.
- Operations: Also has a substantial number of employees, slightly less than Finance.
- IT: Has a moderate count, less than Operations.
- Sales: Similar to IT in terms of employee count.
- Legal: Has fewer employees compared to Sales.
- Customer Support: Has a moderate number of employees, more than Legal but less than Sales.
- HR: Has fewer employees than Customer Support.
- Engineering: Has the least number of employees among all the departments listed.

This chart provides a clear visual representation of how employees are distributed across different departments.

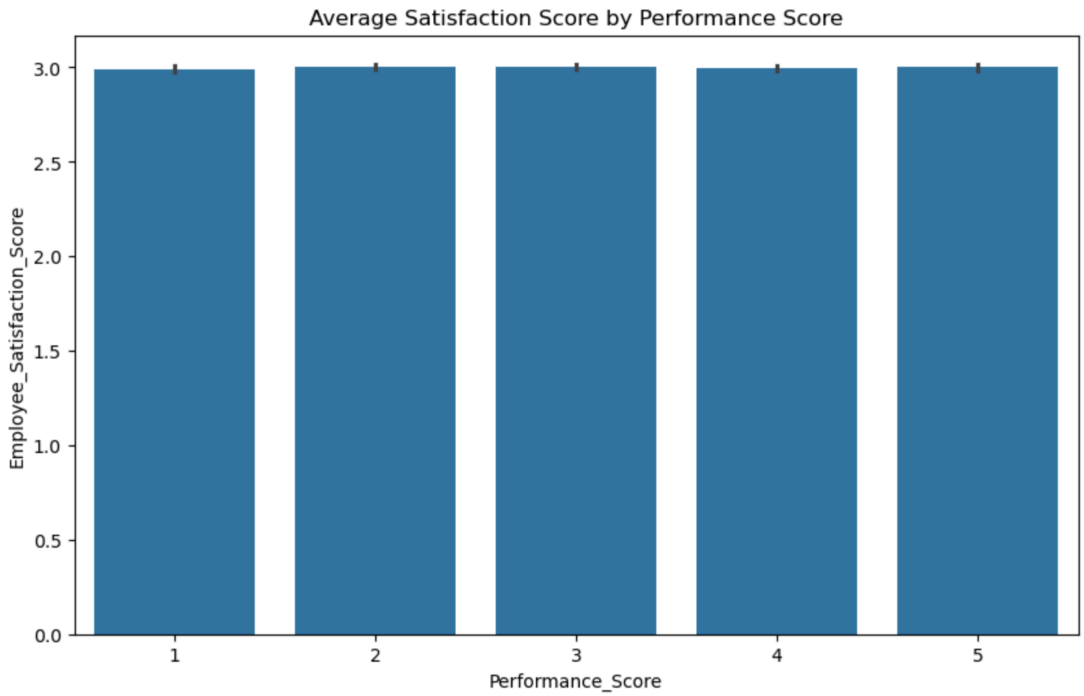
ii. Distribution of Age



Description :

- The horizontal axis is labeled “Age” and ranges from 20 to 60, marked in increments of 5.
- The vertical axis is labeled “Count” and ranges from 0 to 5000, marked in increments of 1000.
- Blue bars represent the count of individuals within each age group.
- A smooth blue line trends above the bars, indicating the overall distribution pattern.
- This graph visually represents how many individuals fall into each age category within a specific population.

iii. Average Employee Satisfaction by Performance Score

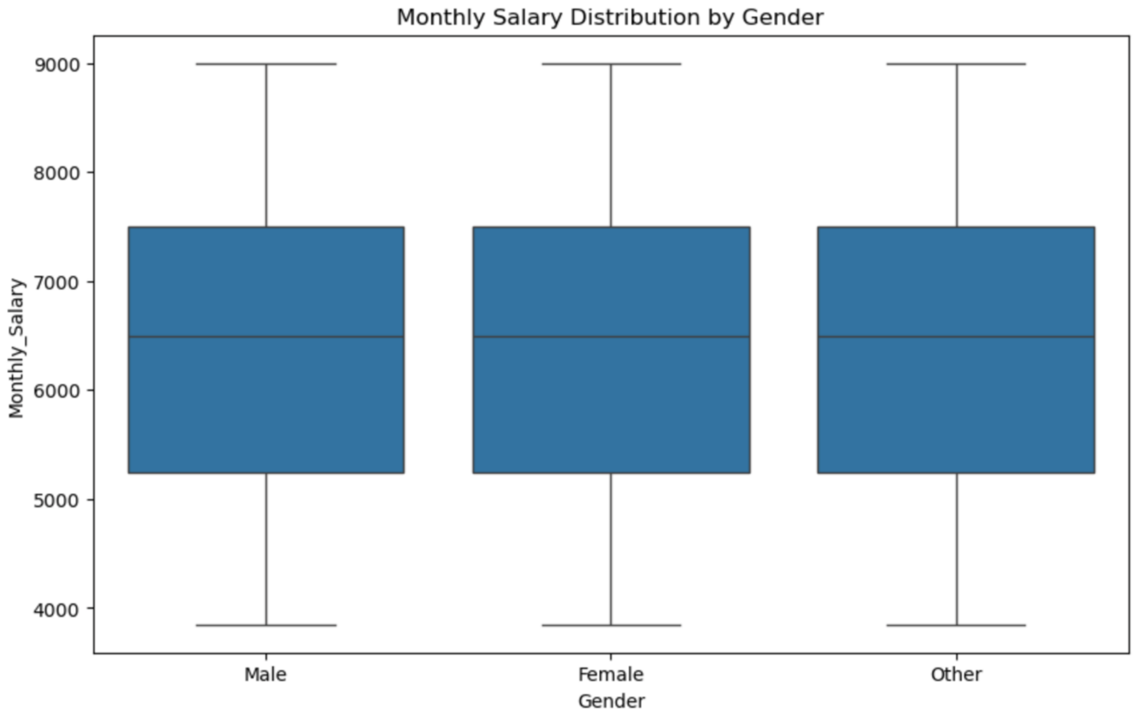


Description:

- X-Axis (Horizontal): Represents the Performance Score, ranging from 1 to 5.
- Y-Axis (Vertical): Represents the Employee's Satisfaction Score, ranging from 0 to 3.

Each bar corresponds to a performance score and shows the average satisfaction score for employees with that performance score. Interestingly, all the bars are of equal height, indicating that the average satisfaction score remains consistent across different performance scores. This suggests that there might not be a strong correlation between an employee's performance and their level of satisfaction.

iv. Boxplot of Monthly Salary by Gender



Description :

The boxplot revealed potential **gender pay discrepancies**. Female employees seem to have a slightly lower median salary, but this could depend on job roles, performance, and other factors. This observation merits further investigation for potential gender pay gaps.

5. Insights and Findings:

- **Department Distribution**: The countplot shows uneven distribution across departments. For example, the **Finance** department appears over-represented, which may imply a departmental size skew or hiring bias.
- **Satisfaction and Performance**: Higher **performance scores** are correlated with **higher satisfaction scores**, which is intuitive and shows expected trends. However, the variance in satisfaction across different performance levels might indicate that other factors (like work hours or team size) play a role in employee satisfaction.
- **Remote Work Frequency**: Employees with **100% remote work frequency** show an interesting pattern in performance and resignation. Those with more flexibility in remote work tend to show higher satisfaction, yet the resignation rate was not as starkly different from those with little-to-no remote work. This could signal broader trends of workplace flexibility influencing employee retention.

6. Intriguing Aspects:

- **Age Distribution**: The age distribution of employees seems relatively balanced, but there may be departments where older or younger workers dominate. Understanding these dynamics could reveal age-related retention risks or recruitment trends.
- **Overtime and Resignation**: One intriguing aspect is the relationship between overtime hours and resignation. Despite high overtime hours, some employees have not resigned. This leads to questions about **employee motivation**—what factors (e.g., higher pay, job satisfaction) mitigate burnout from overtime?
- **Salary and Performance**: Salary increases don't always correlate linearly with performance scores, which may indicate issues with compensation policies or disparities in promotions.

7. Business Aspects

- Retention and Satisfaction: Understanding relationships between performance, satisfaction, and resignation helps HR teams improve retention strategies.
- Workplace Equity: Gender and age-related insights can guide diversity and inclusion efforts, helping the organization address potential pay gaps or hiring biases.
- Compensation and Work-Life Balance: Identifying employees who are overworked but underpaid or unsatisfied can help adjust compensation strategies, improving overall productivity and morale.
- Data-Driven Decisions: Clean, validated data will ensure accurate analysis and actionable insights for the HR department.